



SOFTWARE AND DATABASE DEVELOPMENT LIFECYCLES

IT 525-DATABASE DESIGN AND DATA MODELING

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WHAT IS THE SDLC?

- The Software Development Life Cycle helps organizations enhance their speed, flexibility, communication, and organizational needs by establishing a timeline and strategy for software development and integration into the current workflows.
- This ensures all stakeholders are clear of the expectations, desires, and end results and ensures everyone is clear on their predetermined roles and guidelines for the process to continue seamlessly.
- The seven stages include: planning, analysis, design, coding, testing, deployment, and maintenance.

VARIATION 1-WATERFALL

- Waterfall variation requires each step of the software development lifecycle to be completed prior to continuation of the cycle.
- It is structured and predictable yet inflexible should needs change and can be more time-consuming.
- For well-defined and stable projects, it can provide clear, actionable results for key stakeholders only invested in the main outcomes.

VARIATION 2-AGILE

- Designers make small updates and immediately integrate them coined “sprints” into the final design.
- Allows for more collaboration and input from other stakeholders to continue progressing the design depending on continual input.
- Can address issues before becoming bigger problems.

VARIATION 3-SPIRAL

Four phases

1. Determine goals
 2. Analyze risks and resources
 3. Develop and test new features
 4. Plan for next cycle
- Repeat each phase until project is finished.
 - Ideal for complex projects that require frequent changes.

THE DATABASE DEVELOPMENT LIFECYCLE

1. Database initial study
 2. Database design
 3. Implementation and loading
 4. Testing and evaluation
 5. Operation
 6. Maintenance
- A structured lifecycle utilized to develop strong and effective databases allowing space for proper loading and refining the database throughout the process.

COMPARISON SUMMARY

Lifecycle Type	Definition
Software Development Lifecycle <i>What is the software development lifecycle (SDLC)? n.d.</i>	Strategy for software developers to ensure compliance for coding, testing, and maintenance as the software is developed and utilized.
Variation 1-Waterfall <i>What is the software development lifecycle (SDLC)? n.d.</i>	Requires each step to be fully completed prior to moving on to the next step.
Variation 2- Agile <i>What is the software development lifecycle (SDLC)? n.d.</i>	Collaboration is emphasized during short sprints of development to enhance continuous input from all stakeholders.
Variation 3- Spiral <i>What is the software development lifecycle (SDLC)? n.d.</i>	Establishes four phases to cycle through throughout the lifecycle repeating each cycle as many times as needed to complete the project.
Database Development Lifecycle <i>Database Development Life Cycle, 2025</i>	A lifecycle design specifically for databases focusing on loading data into a database rather than coding systems.

REFERENCES

- What is the software development lifecycle (SDLC)? (n.d.). IBM.
<https://www.ibm.com/think/topics/sdlc>
- Database Development Life Cycle. (2025, August 6). Geeks for Geeks.
<https://www.geeksforgeeks.org/dbms/database-development-life-cycle/>